

System Impact Study Report

for

PJM Generation Interconnection Request
Queue Position AD2-214

“Rock Falls – Garden Plain”

Revision 0: January 2021
Revision 1: June 2022
Revision 2: June 2022
Revision 3: March 2023

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

General

The Interconnection Customer (“IC”) has proposed a 68.0 MW Energy (40.8 MW Capacity) solar PV ground mounted tracking facility to be located in Whiteside County, IL. The IC has proposed an in-service date which was not attainable and will need to get revised during the Facilities Study phase of the project.

Revision History

June 2022 – Revision 1 was issued to reflect a retooled analysis. Revision 2 has been issued to include rating notes.

March 2023 – Revision 3 is issued to include retooled load flow results, stability study results and final Affected System Study results.

Point of Interconnection

The AD2-214 project proposes to interconnect with the ComEd transmission system at a new substation to be constructed adjacent to the Garden Plains-Rock Falls 138kV Line 15518 (see Attachment 1).

Attachment Facilities

The IC AD2-214 generator lead would interconnect to a new 138kV Interconnection Substation. This interconnection would require one 138kV line MOD, a dead-end structure and revenue metering as shown in the one line diagram.

The cost for the attachment facilities is estimated at \$1M.

Scope of Work	Cost Estimate
Installation of one 138kV line MOD, one dead-end structure and one set of revenue metering (see notes below on cost estimate)	\$1,000,000

Direct Connection Network Upgrades

In order to accommodate interconnection of AD2-214, a new 138kV Interconnection Substation would need to be built close to the Garden Plain-Rock Falls 138kV Line 15518 approximately 15.13 miles from the Garden Plain TSS 132.

The scope of work includes the installation of three 138kV circuit breakers in a “breaker-and-a-half” bus configuration and cutting in the Interconnection Substation to Garden Plain-Rock Falls 138kV Line 15518, as shown in the one-line diagram below.

The IC is responsible for constructing all of the facilities on the IC side of the point of interconnection outside of the substation. It is assumed for the purposes of this report that the IC will obtain the site for the Interconnection Substation and right-of-way between the Interconnection Substation and the 138kV transmission line.

In the event that the IC exercises the option to build the interconnecting substation, the IC will be required to construct all interconnection facilities that will be turned over to ComEd in accordance with ComEd published standards and the PJM Tariff.

ComEd would design, engineer and construct the tie in of the Interconnection Substation to the Garden Plain-Rock Falls 138kV Line 15518.

The preliminary cost estimate for Direct Connection Network Upgrade is given in the following tables.

For Option to Build Direct Connection cost estimates:

Scope of Work	Cost Estimate
Installation of a new 138kV substation as described above	N/A
Transmission line tie in work (foundations, structures, conductors)	\$ 2,000,000

ComEd oversight and testing	\$ 1,000,000
Total Cost Estimate (see notes below on cost estimate)	\$ 3,000,000

For ComEd building the interconnecting substation cost estimates:

Scope of Work	Cost Estimate
Installation of a new 138kV substation as described above	\$15,000,000
Transmission line tie in work (foundations, structures, conductors)	\$ 2,000,000
Total Cost Estimate (see notes below on cost estimate)	\$ 17,000,000

ComEd would take approximately 24-months to construct the substation and transmission line work after the ISA / ICSA are signed.

Non-Direct Connection Network Upgrades

The 138kV line 15518 connects to four different substations. Accordingly, integration of the new 138kV Interconnection Substation would require relay/communications/SCADA upgrades at the Garden Plain TSS 132, Nelson TSS 155 and Sterling Steel ESS H71 and Rock Falls TSS 133 substations. The ComEd cost is given below:

Scope of Work	Cost Estimate
Relay/communications/SCADA upgrades at Garden Plain TSS 132	\$ 750,000
Relay/communications/SCADA upgrades at Nelson TSS 155 substation	\$ 750,000
Relay/communications/SCADA upgrades at ESS H71 substation	\$ 750,000
Relay/communications/SCADA upgrades at Rock Falls TSS 133 substation	\$ 750,000
Total Cost Estimate (see notes below on cost estimate)	\$ 3,000,000

Notes on Cost Estimate:

- 1) These estimates are Order-of-Magnitude estimates of the costs that ComEd would bill to the customer for this interconnection. These estimates are based on a one-line electrical diagram of the project and the information provided by the IC.

- 2) There were no site visits performed for these estimates. There may be costs related to specific site related issues that are not identified in these estimates. The site reviews will be performed during the Facilities Study or during detailed engineering.
- 3) These estimates are not a guarantee of the maximum amount payable by the IC and the actual costs of ComEd's work may differ significantly from these estimates. Per the PJM Tariff, IC will be responsible for paying all actual costs of ComEd's work.
- 4) The IC is responsible for all engineering, procurement, testing and construction of all equipment on the IC's side of the Point of Interconnection (POI).
- 5) These cost estimates do not include cost of acquiring right-of-way for the transmission line and purchasing any additional land, if needed, for the line terminations. The need and cost of acquiring property and associated legal costs will be investigated during Facilities Study for this project.

Network Impacts

Queue Project AD2-214 was evaluated as a 68.0 MW (Capacity 40.8 MW) injection on the 138kV line between Rock Falls and Garden Plains (specifically on the ESS H71 – Garden Plains 138 kV line segment) in the ComEd area. Project AD2-214 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AD2-214 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Analysis - 2021

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

Area	Facility Description	Contingency Name	Contingency Type	DC AC	Initial Loading	Final Loading	Rating (MVA)	Rating Type	MVA to Mitigate	MW Contribution
CE	NELSON ; B-ELECT JCT; B 345.0 kV Ckt 1 line	COMED_P4_937-45-BT1-2	Breaker	AC	102.26 %	103.28 %	1837.0	STE	1897.28	18.76
CE	NELSON ; B-ELECT JCT; B 345.0 kV Ckt 1 line	COMED_P4_937-45-BT1-4	Breaker	AC	101.43 %	102.45 %	1837.0	STE	1882.04	18.76

Steady-State Voltage Requirements

(Results of the steady-state voltage studies should be inserted here)

None

Short Circuit

(Summary of impacted circuit breakers)

None

Affected System Analysis & Mitigation

MISO Impacts:

Final results – No Impacts

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

None.

Light Load Analysis - 2021

Not required.

Short Circuit

(Summary form of Cost allocation for breakers will be inserted here if any)

None identified.

Stability and Reactive Power Requirement

There are no mitigations required.

Summer Peak Load Flow Analysis Reinforcements

RTEP ID	Title	Total Cost	Time Estimate
n4734	Electric Junction 345 kV Upgrades	\$4,369,784	24 Months

AD2-214 is a **Contributor**.

Description: 1) Station conductor upgrades (345kV) at Electric Junction. 2) Upgrade 345kV Line & Bus MODs (motor operated disconnect switches at Electric Junction for L15502 Circuit Breaker. Following these upgrades the new rating will be 1529/1765/2041/2428 MVA (SN/SLTE/SSTE/SLD). A preliminary estimate for these upgrades is \$750 K with an estimated construction timeline of 24 months.

Flowgates Addressed by this Reinforcement:

Facility	Contingency
ELECT JCT; B-NELSON ; B 345.0 kV Ckt 1 line	(Any)

New Ratings:

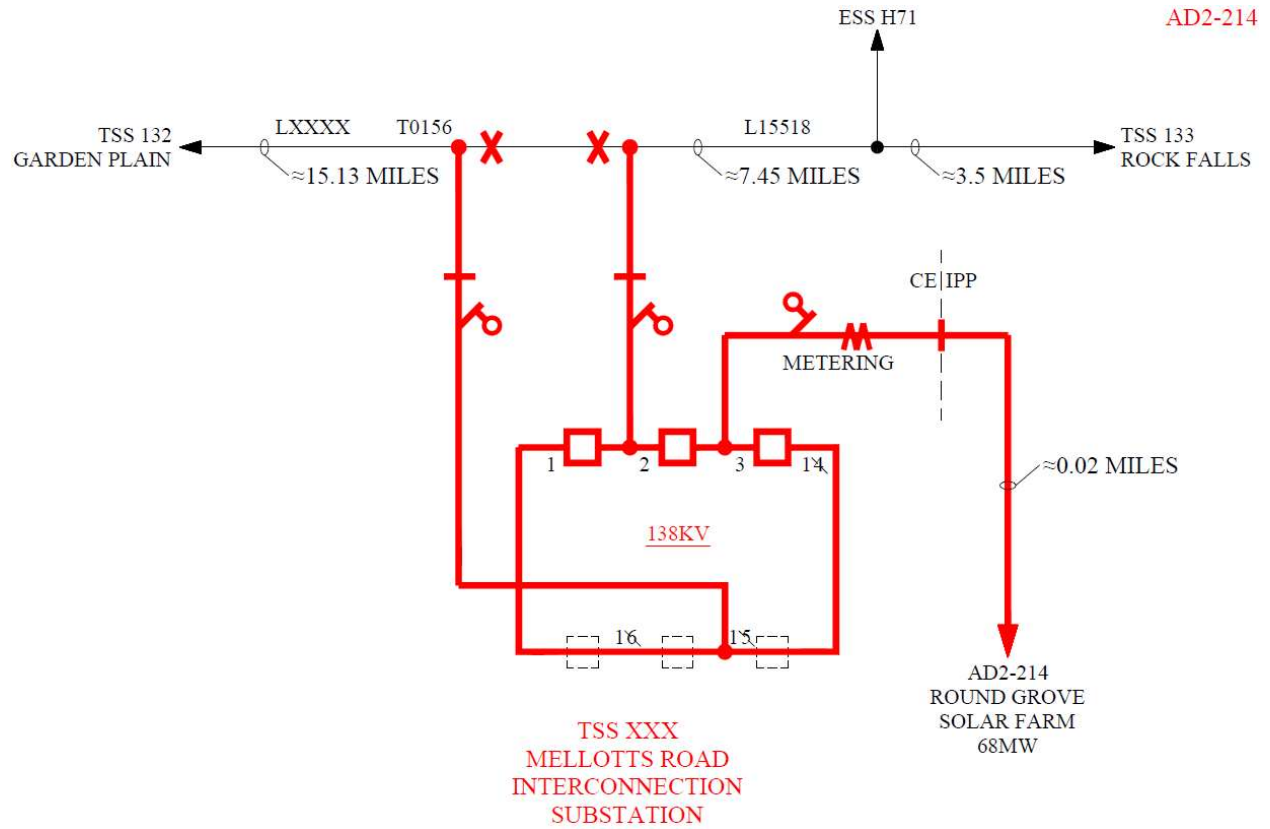
Facility	Rating Set	Rating Type	Rating Value
ELECT JCT; B-NELSON ; B 345.0 kV Ckt 1 line	(All)	A	1529.0 MVA
ELECT JCT; B-NELSON ; B 345.0 kV Ckt 1 line	(All)	B	1765.0 MVA
ELECT JCT; B-NELSON ; B 345.0 kV Ckt 1 line	(All)	C	2428.0 MVA
ELECT JCT; B-NELSON ; B 345.0 kV Ckt 1 line	(All)	STE	2041.0 MVA

Cost Allocation:

Queue Project	MW Impact	Percent Allocation	Allocated Cost (\$USD)
AD2-214	18.8 MW	100.0%	\$4,369,784

Note: AD2-214 is the first project to meet cost allocation requirements and thus is allocated 100% of the costs for this upgrade.

ATTACHMENT 1



Details for NELSON ; B-ELECT JCT; B 345.0 kV Ckt 1 line l/o COMED_P4_937-45-BT1-2__

Bus #	Bus Name	Type	Full MW Contribution	GenDeliv MW Contribution
290051	GSG-6; E	Adder	8.94 MW	7.6 MW
295109	WESTBROOK E	Adder	4.79 MW	4.07 MW
295111	SUBLETTE E	Adder	2.33 MW	1.98 MW
926331	AC1-110 1	Adder	-1.05 MW	-0.89 MW
926341	AC1-110 2	Adder	-1.05 MW	-0.89 MW
934051	AD1-031 C O1	Adder	3.05 MW	2.59 MW
934052	AD1-031 E O1	Adder	4.98 MW	4.23 MW
934431	AD1-067 C	Adder	0.11 MW	0.1 MW
934432	AD1-067 E	Adder	0.47 MW	0.4 MW
934701	AD1-098 C O1	Adder	6.31 MW	5.37 MW
934702	AD1-098 E O1	Adder	4.61 MW	3.92 MW
937001	AD2-134 C	Adder	2.16 MW	1.84 MW
937311	AD2-172 C	Adder	1.59 MW	1.35 MW
937312	AD2-172 E	Adder	2.2 MW	1.87 MW
937531	AD2-214 C	50/50	11.26 MW	11.26 MW
937532	AD2-214 E	50/50	7.5 MW	7.5 MW
274700	CORDOVA ;2C	50/50	7.21 MW	7.21 MW
274701	CORDOVA ;1S	50/50	8.11 MW	8.11 MW
274715	NELSON EC;1C	50/50	7.82 MW	7.82 MW
274716	NELSON EC;1S	50/50	10.8 MW	10.8 MW
274717	NELSON EC;2C	50/50	7.82 MW	7.82 MW
274718	NELSON EC;2S	50/50	10.8 MW	10.8 MW
274849	CRESCENT ;1U	Adder	4.25 MW	3.61 MW
274850	MENDOTA H;RU	Adder	4.62 MW	3.92 MW
293516	O-009 E1	Adder	19.72 MW	16.76 MW

293517	O-009 E2	Adder	10.01 MW	8.51 MW
293518	O-009 E3	Adder	11.03 MW	9.37 MW
293715	O-029 E	Adder	15.04 MW	12.78 MW
293716	O-029 E	Adder	11.53 MW	9.8 MW
293717	O-029 E	Adder	10.6 MW	9.01 MW
293771	O-035 E	Adder	5.62 MW	4.78 MW
294401	BSHIL;1U E	Adder	9.22 MW	7.84 MW
294410	BSHIL;2U E	Adder	9.22 MW	7.84 MW
294763	P-046 E	Adder	6.07 MW	5.16 MW
925581	AC1-033 C	Adder	1.51 MW	1.28 MW
925582	AC1-033 E	Adder	10.11 MW	8.6 MW
990901	L-005 E	Adder	12.77 MW	10.86 MW
919581	AA2-030	50/50	54.6 MW	54.6 MW
919221	AA1-146	50/50	54.6 MW	54.6 MW
925302	AB2-191 E	Adder	1.18 MW	1.01 MW
925301	AB2-191 C	Adder	0.86 MW	0.73 MW
927201	AC1-214 C	Adder	1.81 MW	1.53 MW
927202	AC1-214 E	Adder	5.74 MW	4.88 MW
933911	AD1-013 C O1	Adder	1.51 MW	1.28 MW
933912	AD1-013 E O1	Adder	2.41 MW	2.05 MW
938122	J836 E	External Queue	14.96 MW	14.96 MW
939221	J877	External Queue	18.53 MW	18.53 MW
939391	J898 C	External Queue	1.47 MW	1.47 MW
939392	J898 E	External Queue	5.87 MW	5.87 MW
939751	J844 C	External Queue	5.8 MW	5.8 MW
939752	J844 E	External Queue	23.18 MW	23.18 MW

940181	J407 C	External Queue	3.09 MW	3.09 MW
940182	J407 E	External Queue	12.36 MW	12.36 MW
940211	J411 C	External Queue	4.12 MW	4.12 MW
940212	J411 E	External Queue	16.46 MW	16.46 MW
940221	J416 C	External Queue	3.94 MW	3.94 MW
940222	J416 E	External Queue	15.76 MW	15.76 MW
940401	J041 C	External Queue	1.67 MW	1.67 MW
940402	J041 E	External Queue	6.7 MW	6.7 MW
940441	J412 C	External Queue	2.78 MW	2.78 MW
940442	J412 E	External Queue	11.13 MW	11.13 MW
940471	J438 C	External Queue	3.47 MW	3.47 MW
940472	J438 E	External Queue	13.86 MW	13.86 MW
940491	J443 C	External Queue	0.97 MW	0.97 MW
940492	J443 E	External Queue	3.89 MW	3.89 MW
940501	J449 C	External Queue	3.28 MW	3.28 MW
940502	J449 E	External Queue	13.13 MW	13.13 MW
940522	J455 E	External Queue	22.24 MW	22.24 MW
951211	J512 C	External Queue	3.25 MW	3.25 MW
951212	J512 E	External Queue	13.0 MW	13.0 MW
951231	J541 C	External Queue	5.81 MW	5.81 MW
951232	J541 E	External Queue	23.24 MW	23.24 MW
951251	J555 C	External Queue	3.78 MW	3.78 MW
951252	J555 E	External Queue	15.14 MW	15.14 MW
951421	J614 C	External Queue	0.98 MW	0.98 MW
951422	J614 E	External Queue	3.91 MW	3.91 MW

951581	J583 C	External Queue	2.8 MW	2.8 MW
951582	J583 E	External Queue	11.19 MW	11.19 MW
951601	J590 C	External Queue	1.23 MW	1.23 MW
951602	J590 E	External Queue	4.91 MW	4.91 MW
960000	G735 E	External Queue	13.2 MW	13.2 MW
960001	G798 E	External Queue	10.54 MW	10.54 MW
960004	G870 E	External Queue	12.04 MW	12.04 MW
960005	G947 E	External Queue	7.12 MW	7.12 MW
960006	H008 E	External Queue	2.92 MW	2.92 MW
960007	H021 E	External Queue	10.32 MW	10.32 MW
960008	H096 E	External Queue	3.04 MW	3.04 MW
960009	J091 E	External Queue	4.36 MW	4.36 MW
960012	J191 E	External Queue	5.43 MW	5.43 MW
960014	R42 E	External Queue	6.73 MW	6.73 MW
960015	R42 E2	External Queue	8.03 MW	8.03 MW
960019	J278 E	External Queue	5.64 MW	5.64 MW
960020	J278 E2	External Queue	5.64 MW	5.64 MW
970031	J475 C	External Queue	3.78 MW	3.78 MW
970032	J475 E	External Queue	15.14 MW	15.14 MW
970111	J495 C	External Queue	2.89 MW	2.89 MW
970112	J495 E	External Queue	11.55 MW	11.55 MW
970141	J498 C	External Queue	5.08 MW	5.08 MW
970142	J498 E	External Queue	20.33 MW	20.33 MW
970151	J499 C	External Queue	4.95 MW	4.95 MW
970152	J499 E	External Queue	19.8 MW	19.8 MW

970161	J500 C	External Queue	6.88 MW	6.88 MW
970162	J500 E	External Queue	27.52 MW	27.52 MW
970191	J504	External Queue	5.87 MW	5.87 MW
970212	J506 E	External Queue	13.32 MW	13.32 MW
970231	J514	External Queue	3.5 MW	3.5 MW
970251	J523	External Queue	2.24 MW	2.24 MW
970261	J524	External Queue	4.61 MW	4.61 MW
970291	J527 C	External Queue	3.42 MW	3.42 MW
970292	J527 E	External Queue	13.69 MW	13.69 MW
970311	J529 C	External Queue	3.55 MW	3.55 MW
970312	J529 E	External Queue	14.19 MW	14.19 MW
970321	J530 C	External Queue	5.59 MW	5.59 MW
970322	J530 E	External Queue	22.35 MW	22.35 MW
970351	J534 C	External Queue	3.71 MW	3.71 MW
970352	J534 E	External Queue	14.82 MW	14.82 MW
970361	J535 C	External Queue	3.1 MW	3.1 MW
970362	J535 E	External Queue	12.4 MW	12.4 MW
981491	J748 C	External Queue	2.72 MW	2.72 MW
981492	J748 E	External Queue	10.87 MW	10.87 MW
983001	J285 C	External Queue	3.34 MW	3.34 MW
983002	J285 E	External Queue	10.01 MW	10.01 MW
983011	J289 C	External Queue	0.29 MW	0.29 MW
983012	J289 E	External Queue	0.87 MW	0.87 MW
983061	J329	External Queue	4.72 MW	4.72 MW
983121	J343 C	External Queue	2.02 MW	2.02 MW
983122	J343 E	External Queue	6.06 MW	6.06 MW

LGEE	LTFEXP_LGEE->LTFIMP_LGEE	CLTF	0.06 MW	0.06 MW
CIN	LTFEXP_CIN->LTFIMP_CIN	CLTF	0.49 MW	0.49 MW
CPLE	LTFEXP_CPLE->LTFIMP_CPLE	CLTF	0.35 MW	0.35 MW
IPL	LTFEXP_IPL->LTFIMP_IPL	CLTF	0.25 MW	0.25 MW
CBM-W2	LTFEXP_CBM-W2->LTFIMP_CBM-W2	CLTF	48.07 MW	48.07 MW
CARR	LTFEXP_CARR->LTFIMP_CARR	CLTF	0.08 MW	0.08 MW
CBM-W1	LTFEXP_CBM-W1->LTFIMP_CBM-W1	CLTF	43.17 MW	43.17 MW
O-066	LTFEXP_O-066->LTFIMP_O-066	CLTF	0.96 MW	0.96 MW
CBM-S2	LTFEXP_CBM-S2->LTFIMP_CBM-S2	CLTF	1.91 MW	1.91 MW
CBM-S1	LTFEXP_CBM-S1->LTFIMP_CBM-S1	CLTF	4.7 MW	4.7 MW
G-007	LTFEXP_G-007->LTFIMP_G-007	CLTF	0.15 MW	0.15 MW
MEC	LTFEXP_MEC->LTFIMP_MEC	CLTF	53.98 MW	53.98 MW
CLIFTY	LTFEXP_CLIFTY->LTFIMP_CLIFTY	CLTF	1.22 MW	1.22 MW